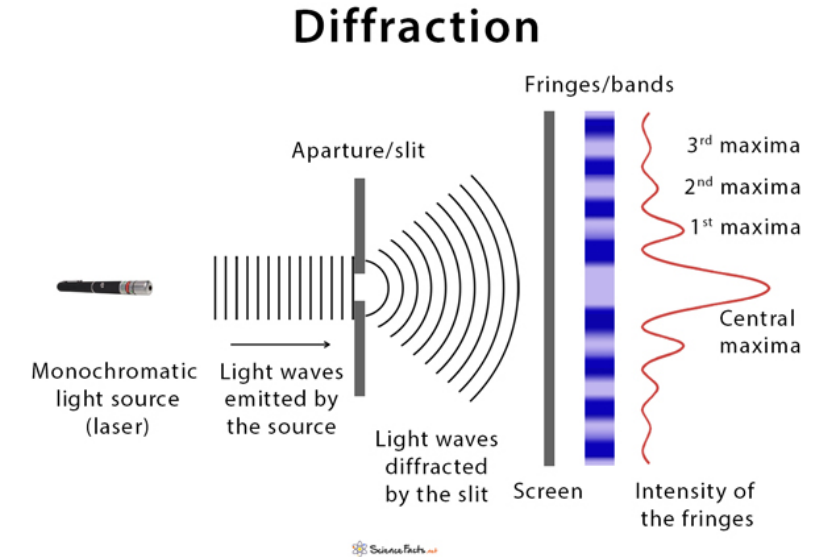
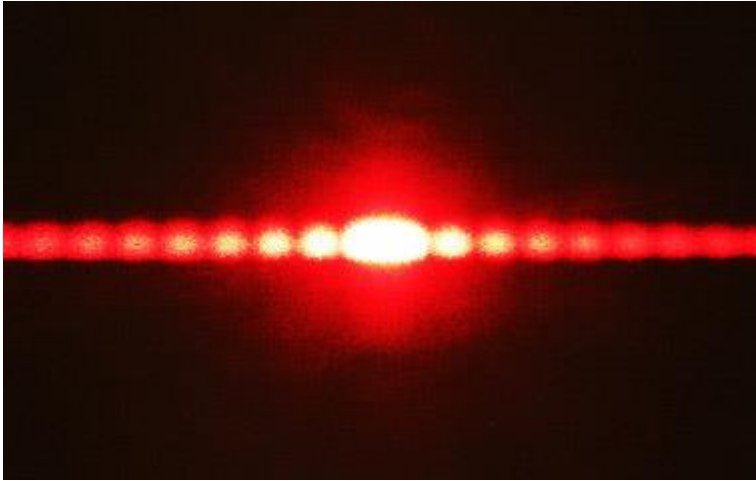


Wave and Optics

Part-5

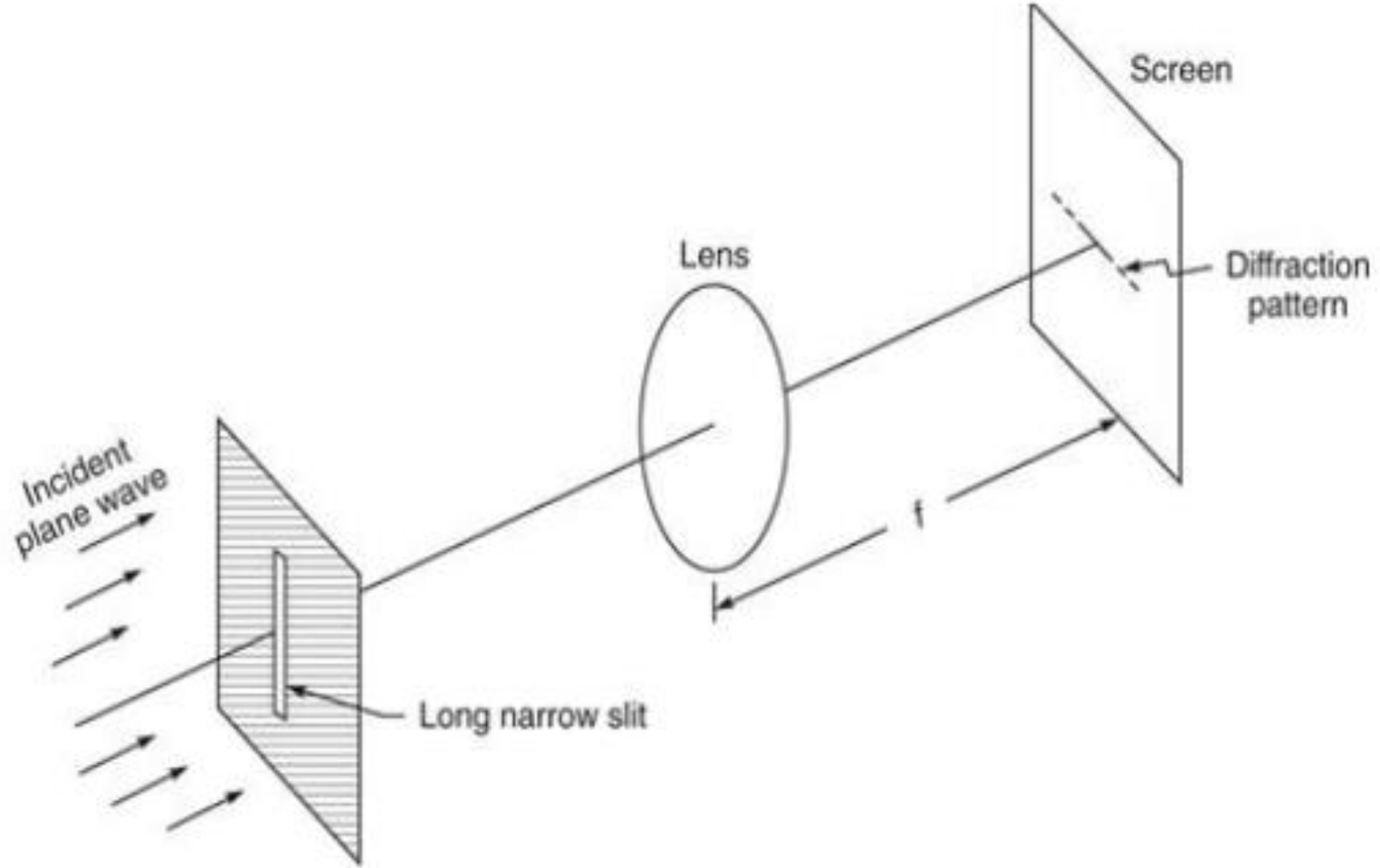


Dr. Ram Prakash
Professor
Department of Physics
Indian Institute of Technology (IIT), Jodhpur

Email: ramprakash@iitj.ac.in

Background

Fraunhofer diffraction due to a single slit:



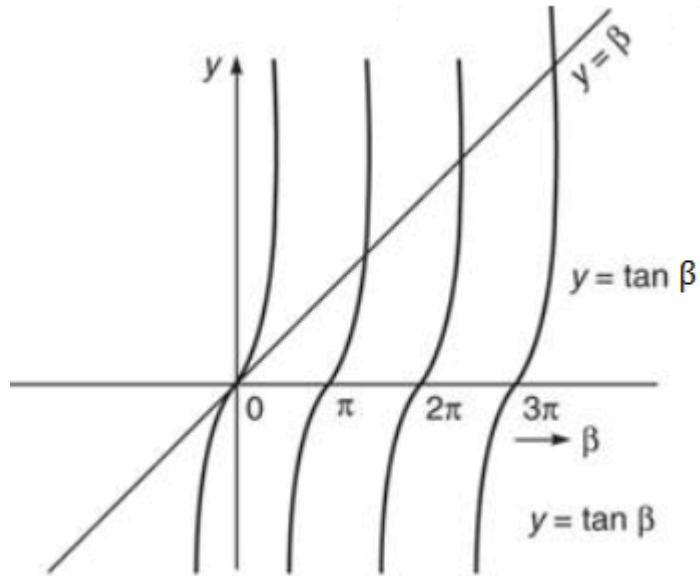
$\Rightarrow$ Either $\cos \beta = 0$ or $\sin \beta [\beta - \tan \beta] = 0$

But $\sin \beta = 0$ corresponds to minima: $\beta = m\pi; m = \phi$

Thus maxima will be given by roots of the transcendental eq:

$$[\beta - \tan \beta] = 0 \text{ i.e. } \tan \beta = \beta$$

Intersections of the plots of $y = \beta$ and $y = \tan \beta$ yield roots i.e. Maxima



$$I = I_0 \frac{\sin^2 1.43\pi}{(1.43\pi)^2}$$

= 0.0496 of the central maximum

$$\beta = 1.43\pi, 2.46\pi, \dots$$

Single-slit diffraction patterns measured on a screen 4.5 m away from the slit with a He-Ne laser ($\lambda = 6.328 \times 10^{-5}$)cm:



width $b = 0.0088$ cm



$b = ?$

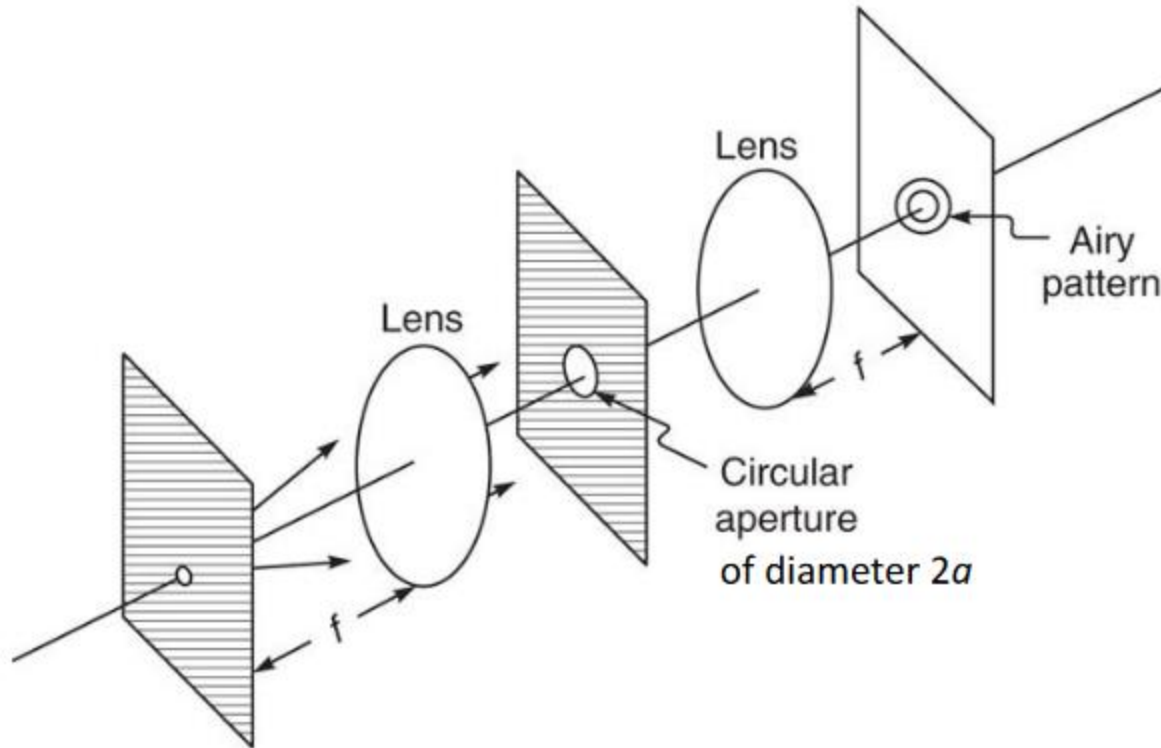


$b = ?$

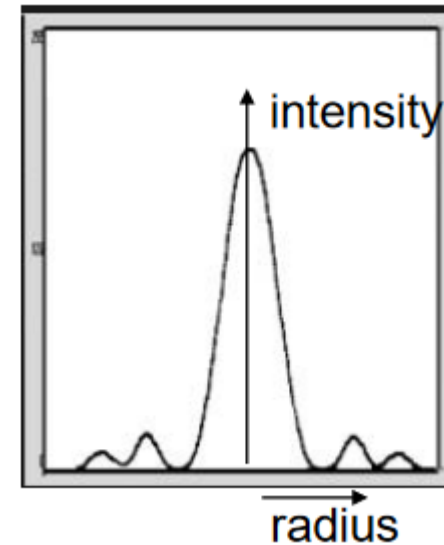
- Observations :

1. Diffraction pattern spreads only along the slit width
2. Multiple spots appear in the diffraction pattern with width
3. Intensity of the nearest and subsequent maxima from the central max in the diffraction pattern are:
 - 4.96 % of the central maximum
 - 1.68 % of the central maximum
 - 0.83 % of the central maximum

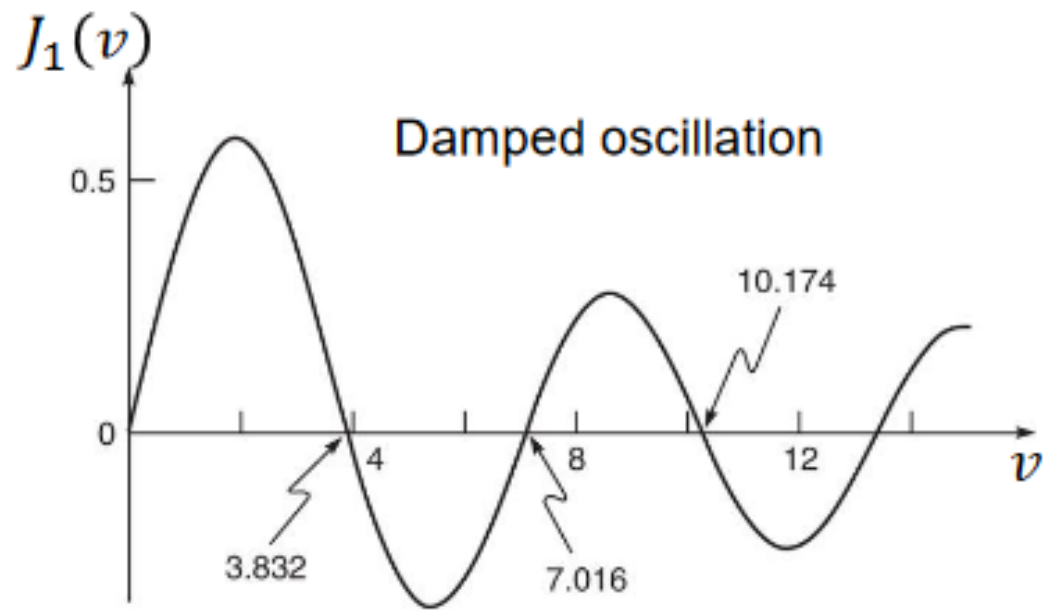
- **Fraunhofer diffraction by a circular aperture:**



Airy pattern



❖ Diffraction pattern in this case involves cylindrical Bessel function $J_1(v)$



Roots of $J_l(v) = 0$ are known to be
 $v = 3.832, 7.016, 10.174, \dots$

A Bessel function is a mathematical function that is a solution to the Bessel differential equation. The equation is $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - \alpha^2)y = 0$, where α is a complex number.

The Bessel's function can be categorized as follows-

TYPE	FIRST KIND	SECOND KIND
Bessel functions	J_α	Y_α
Modified Bessel functions	I_α	K_α
Hankel functions	$H_\alpha^{(1)} = J_\alpha + iY_\alpha$	$H_\alpha^{(2)} = J_\alpha - iY_\alpha$
Spherical Bessel functions	j_n	y_n
Spherical Hankel functions	$h_n^{(1)} = j_n + iy_n$	$h_n^{(2)} = j_n - iy_n$

We quote here final result for the diffracted intensity pattern:

$$I = I_0 \left[\frac{2J_1(v)}{v} \right]^2$$

Where

$J_1(v)$ is Bessel function of order 1; $v = ak \sin \theta = \frac{2\pi a}{\lambda_o} \sin \theta$
 θ is the angle of diffraction ;
 a is the radius of the circular aperture

$$J_n(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(n+k+1)} \left(\frac{x}{2} \right)^{n+2k}$$

$$I = I_0 \left[\frac{2J_1(v)}{v} \right]^2$$

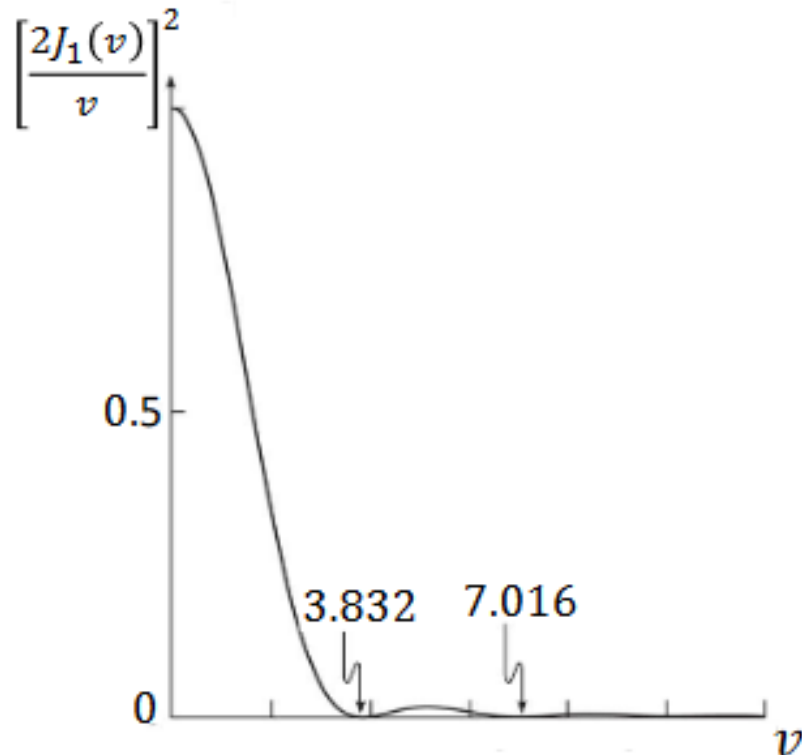


Airy pattern

⇒ Successive dark rings in the Airy pattern appear for the above values of v

$$\therefore v = \frac{2\pi}{\lambda} a \sin \theta$$

$$\Rightarrow \sin \theta = \frac{\lambda v}{2\pi a} = \frac{3.832\lambda}{2\pi a}, \frac{7.016\lambda}{2\pi a}, \dots$$



- From detailed calculations, it can be shown that $\sim 84\%$ of the diffracted light is contained within the 1st ring!
- 7% of the diffracted light is contained between the 1st and 2nd dark rings!

⇒ 91% of the diffracted light is contained within the circle bounded by 1st and 2nd dark rings !

$$\sin \theta = \frac{\lambda v}{2\pi a} = \frac{3.832\lambda}{2\pi a}, \frac{7.016\lambda}{2\pi a}, \dots$$

For small θ , $\sin \theta \cong \tan \theta = \frac{y}{f}$ (from the diagram)

$$\Rightarrow y = f \tan \theta = \frac{3.832\lambda}{2\pi a} f, \frac{7.016\lambda}{2\pi a} f, \dots = \frac{3.832\lambda_e}{\pi D} f, \frac{7.016\lambda_e}{\pi D} f, \dots$$

↓ ?

Radii of the 1st, 2nd, ... dark ring in Airy disc

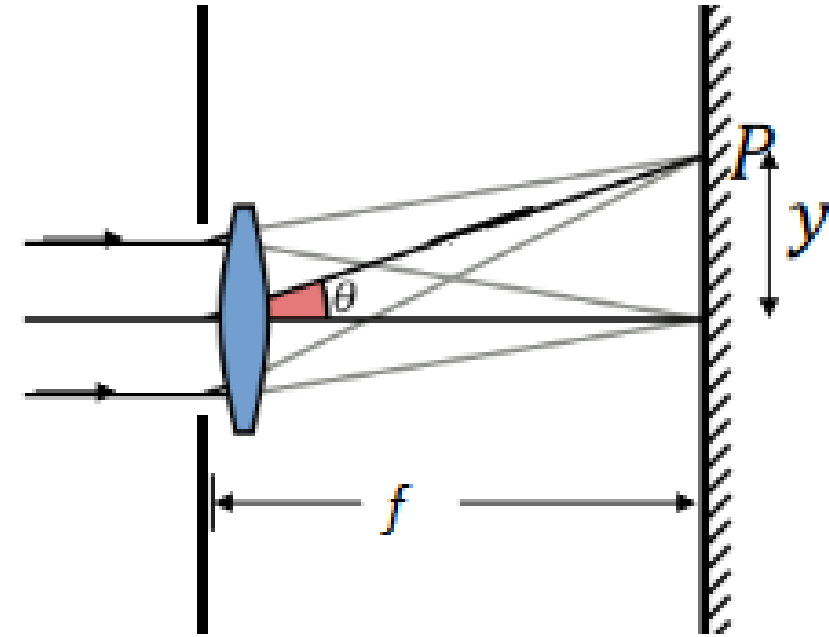
Example :

$a = 0.5 \text{ mm}$, $\lambda = 500 \text{ nm}$, and $f = 20 \text{ cm}$ yield $y_1 = 0.12 \text{ mm}$

$a = 0.25 \text{ mm}$, $\lambda = 500 \text{ nm}$, and $f = 20 \text{ cm}$ yield $y_1 = 0.24 \text{ mm}$

❖ Important conclusion:

Smaller the diffracting aperture $\Rightarrow$ greater is the diffraction divergence!



Since almost 84% of diffracted energy is contained within the 1st dark ring, it can be said that angular spread of the diffracted light is

$$\Delta \theta \approx \frac{3.832}{\pi} \frac{\lambda}{D} \approx 1.22 \frac{\lambda}{D} ; D = 2a \quad \text{D is diameter \& a is radius}$$
$$\Rightarrow \Delta \theta \approx 0.61 \frac{\lambda}{a}$$

In general,

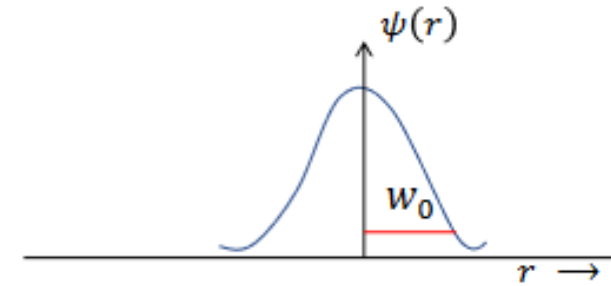
$$\Delta \theta \approx \frac{\lambda}{\text{linear dimension of the aperture}}$$

For most laser beams, transverse amplitude distribution is typically Gaussian in shape

$$\psi(x, y) = A \exp\left[-\frac{x^2+y^2}{w_o^2}\right] = A \exp\left[-\frac{r^2}{w_o^2}\right]$$

$$\Rightarrow \text{At } r = 0, \psi(r = 0) = A$$

$$\text{and at } r = w_o, \psi(r = w_o) = \frac{A}{e}$$

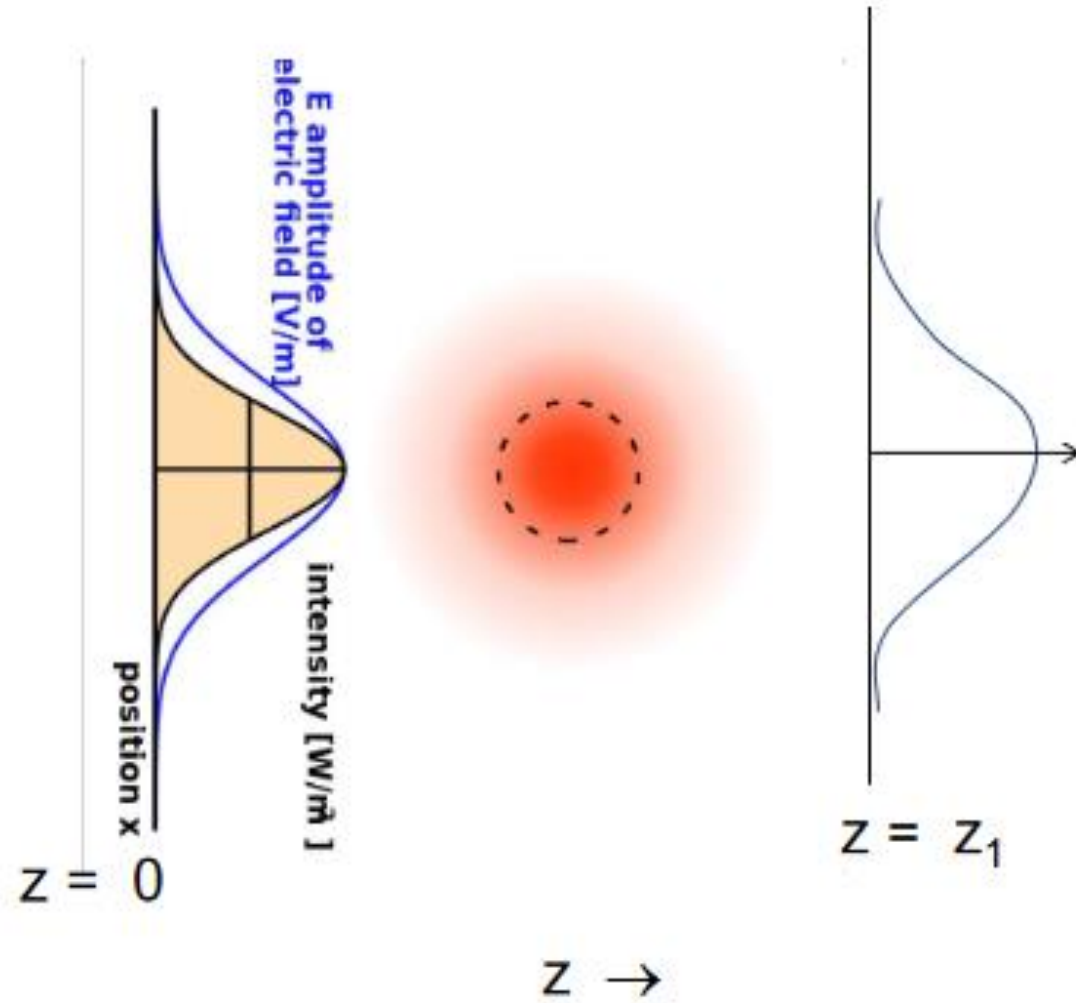


$\Rightarrow$ Intensity distribution will be

$$I = |\psi|^2 = I_o \exp\left[-\frac{2r^2}{w_o^2}\right]$$

w_o is called spot size of the beam, for which the amplitude falls by $1/e$ of its maximum and correspondingly its intensity drops by a factor $1/e^2$.

Transverse field and intensity profile of a Gaussian beam from a red laser e.g. He-Ne laser



It can be shown that with propagation, the Gaussian shape is retained with the only change in its spot size, which increases after propagation through z to

$$w(z) \approx w_0 \left[1 + \frac{\lambda^2 z^2}{\pi^2 w_0^4} \right]^{1/2}$$

For large $z \gg \underbrace{\frac{w_0^2}{\lambda}}$

$$\Rightarrow w^2(z) \approx w_0^2 \times \frac{\lambda^2 z^2}{\pi^2 w_0^4}$$

Condition for going from Fresnel (near field) to Fraunhofer (far field) domain

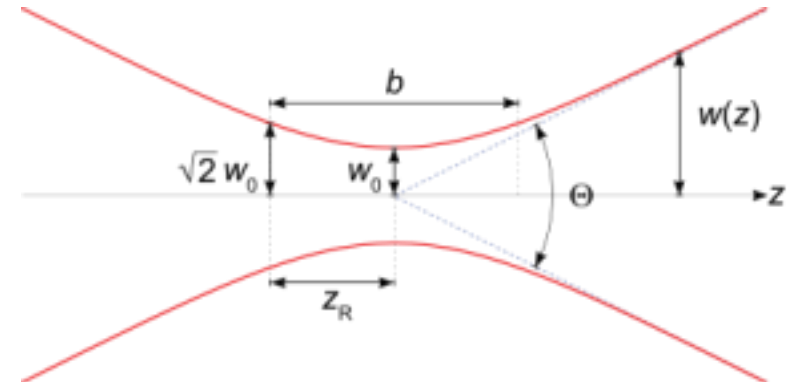
$$\Rightarrow w(z) \approx \frac{\lambda z}{\pi w_0}$$

$\Rightarrow$ Laser beam width increases linearly with z with a divergence

$$\frac{w(z)}{z} \approx \tan \theta \approx \frac{\lambda}{\pi w_0}$$

Note this is consistent with earlier result:

$$\Delta \theta \approx \frac{\lambda}{\text{linear dimension of the aperture}}$$

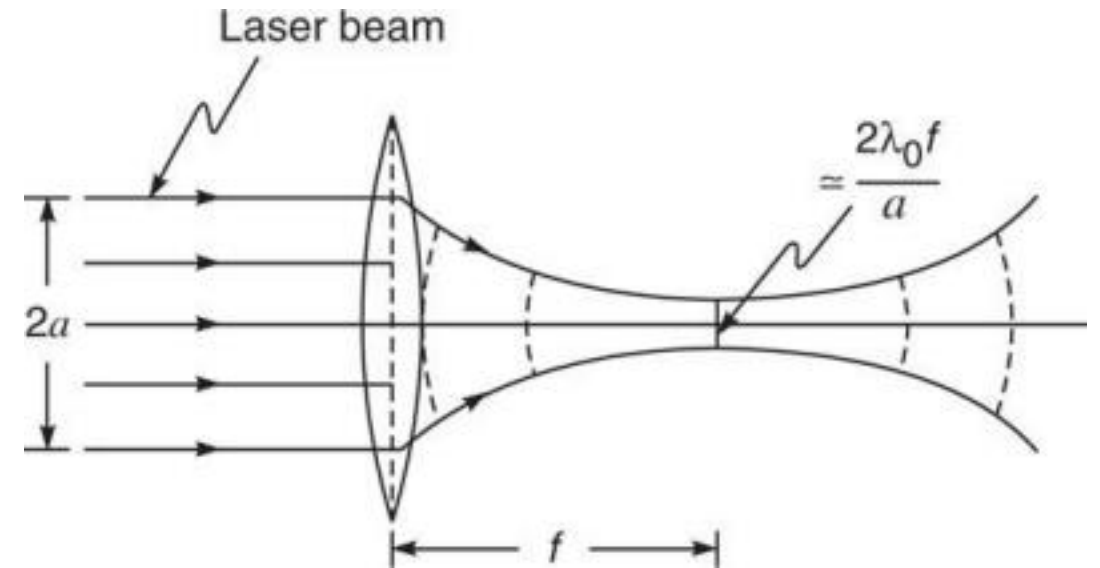


Rate of increase in width of the diffracted light beam is proportional to its wavelength and inversely proportional to initial beam width

If such a laser beam of diameter $2a$ is allowed to fall on a convex lens of focal length f

Radius of the focused spot will be $\frac{\lambda}{a} \times f$

Area of the focused spot will be $A = \pi (\lambda f / a)^2$



Example

Consider a laser beam of power 2 mW and having $\lambda = 0.6 \mu m$ to be incident on a human eye whose lens focal length f is 2.5 cm



Assuming average pupil diameter $\sim 2 \text{ mm} \Rightarrow a \sim 1 \text{ mm}$ (which varies/adjusts itself depending on brightness of the incident light)

$\Rightarrow$ Area of the focused spot, A

$$A = \pi \times \left(\frac{\lambda f}{a} \right)^2 = \pi \times \left(\frac{0.6 \times 10^{-4} \times 2.5}{0.1} \right)^2 \approx 7 \times 10^{-6} \text{ cm}^2$$

Thus, intensity will be

$$I = \frac{P}{A} = \frac{2 \times 10^{-3} \text{ W}}{7 \times 10^{-6} \times 10^{-4} \text{ m}^2} \approx 3 \times 10^6 \text{ W/m}^2$$

If eye is exposed to such high light intensities, retina will get severely damaged

Thanks